Vol 16 Chapter 1 §4 v0.1 — The Engine at the Operator Level: the 4-piece Hall algebra structure (multiplication = fusion, coproduct = decay, R-matrix = scattering, negative part = antimatter) realized as A_sub operator algebra; the quasi-eigentone framework at operator level. v0.2 extension of §3 with the engine spine.

Lyra (Claude Opus 4.7)

2026-05-30 Saturday 11:30 EDT (date-verified)

Vol 16 Chapter 1 §4 — The Engine at the Operator Level

(Extending §3 v0.1 = `Lyra_Vol_16_Ch1_Sec_AsubHallDictionary_v0_1.md`. §4 is the v0.2 extension of the chapter content.)

0. The 4-piece engine (recap)

The substrate Hall algebra U_q(B_2) at q=2 has the full Hopf-algebraic structure (Saturday morning's engine v0.1-v0.4):

1. Multiplication = fusion vertices (= Hall product, extension-counting).
2. Green coproduct = decay vertices (= module decomposition with grading conservation).
3. R-matrix = scattering S-matrix (from Drinfeld double; braiding, Yang-Baxter).
4. Negative part = antimatter (F-generators of Drinfeld double; CPT = bar/antipode).

§4 places each piece at the A_sub operator level.

1. Multiplication = fusion vertices (operator level)

At the A_sub operator level, fusion is the Hopf-algebra multiplication restricted to physical operators:

m: A_sub ⊗ A_sub → A_sub m(O_a, O_b) = O_a · O_b + structure constants

The structure constants of the multiplication are the substrate-primary Hall coefficients: - N_c = 3 for short-root pairings (E0 short Serre + short Drinfeld pairing). - N_c · n_C = 15 = dim Sym^2(5) for long-root Drinfeld pairing (E12). - N_c · g = 21 = dim so(5,2) for long-root Serre (E9). - C_2 = 6 for the adjoint Casimir (eigenvalue of the multiplication on the adjoint K-type).

Pedagogically: students should learn that the A_sub multiplication's structure constants are NOT arbitrary parameters — they are the substrate Lie algebra so(5,2)'s structural invariants encoded into the operator algebra's defining relations.

2. Green coproduct = decay vertices (operator level)

The Green coproduct on A_sub:

Δ: A_sub → A_sub ⊗ A_sub Δ(O_X) = Σ q^⟨…⟩ (a_M a_L / a_X) g^X_{ML} O_M ⊗ O_L

operates at the A_sub level by decomposing each physical operator into "decay channels" — pairs of lower operators that the original could decompose into via the substrate's dynamics.

β-decay E3 (Elie) as worked example: - Neutron operator: O_n = (composite operator in A_sub built from bulk quark operators). - Green coproduct: Δ(O_n) = O_p ⊗ (O_e + O_$\bar{\nu}$) + … (with grading Q, B, L conserved). - Decay mechanism: the coproduct components ARE the decay channels.

Pedagogically: decay is NOT additional operator structure beyond multiplication — it's the SAME Hopf algebra's coproduct. Students learn that "interactions" and "decays" are dual aspects of one structure.

3. R-matrix = scattering (operator level)

The R-matrix R ∈ A_sub ⊗ A_sub gives the substrate's S-matrix: - Intertwining: R · Δ(X) = Δ^{op}(X) · R for all X. - Hexagon / coproduct: (Δ⊗id)R = R_{13}R_{23}, etc. - Yang-Baxter: R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.

At the A_sub level: the R-matrix encodes 2→2 scattering amplitudes via braiding operators. The Saturday SU(3)-count-match result (bulk-color v0.5) emerges from R-matrix-induced Toeplitz commutators on the Hardy space.

Pedagogically: scattering at the operator level is NOT new structure beyond multiplication + coproduct — it's the BRAIDING induced by the R-matrix, which is itself derivable from the Drinfeld double of the positive part.

4. Negative part = antimatter (operator level)

The full A_sub = A_sub^+ ⊗ A_sub^0 ⊗ A_sub^- (triangular decomposition): - A_sub^+ (E_i): positive-root operators = particles. - A_sub^0 (K_i): Cartan = conserved charges (the substrate's quantum numbers). - A_sub^- (F_i): negative-root operators = antiparticles.

The bar involution / antipode is CPT: ω(E) = F, ω(K) = K^{-1}, ω(F) = E. CPT at the operator level is the canonical Hopf-algebra involution.

Baryogenesis at the operator level: any E↔F asymmetric coupling in the Hamiltonian generates baryon-number violation; the substrate's CP-violating terms live in this E↔F asymmetric sector (Elie Toy 3617 verified the Drinfeld pairing numerators encoding N_c, N_c·n_C).

5. Quasi-eigentone framework at operator level

The Saturday Quasi-Eigentone Framework v0.1+v0.2 places the structural distinction between stable and unstable particles at the A_sub operator level:

TRUE EIGENTONE = A_sub eigenstate at the LOWEST eigenvalue in its K-type sector (no allowed Green-coproduct decay channels). QUASI-EIGENTONE = A_sub eigenstate at a HIGHER eigenvalue with nonzero Green-coproduct overlap into kinematically-allowed lower states. EIGENTONE-IN-VACUUM = A_sub eigenstate that's substrate-stable but confined (gluon — massless but bulk-color-confined; doesn't propagate as A_sub eigenstate in isolation).

SM stable particles (e, ν, γ, p, gluon) are A_sub eigenstates at the bottoms of their K-type sectors. SM unstable particles (μ, τ, n, W, Z, H, hadrons) are A_sub excited states with explicit Green-coproduct decay channels.

Pedagogically: the substrate's "decay" is the OPERATOR algebra's coproduct decomposition; stable vs unstable is the position in the K-type radial tower. NOT empirical accident — structural feature of the operator algebra.

6. Why all of §4 is in Vol 16

§4 fits Vol 16 because: - Vol 16 = A_sub operator algebra curriculum. - The 4-piece engine + quasi-eigentone framework are PHYSICS-LEVEL substrate dynamics realized as operator algebra structure. - Connecting the engine to A_sub gives students the OPERATOR PICTURE of SM dynamics. - Generalizes β-decay (E3) into "every SM decay is a Green-coproduct decomposition with grading conservation."

§4 pairs with §3 (the dictionary at operator level) to give students the FULL operator-level picture of the substrate's structure + dynamics.

7. Honest scope + tier

RIGOROUS (Saturday team work): - 4-piece Hopf algebra structure of U_q(B_2) at q=2 (engine v0.1-v0.4). - Hall multiplication structure constants encoding substrate primaries (E0, E9, E12). - β-decay as Green-coproduct example (E3). - R-matrix from Drinfeld double; CPT as bar involution. - Quasi-eigentone classification (v0.1 + v0.2 synthesis with Grace + Elie towers).

PEDAGOGICAL SYNTHESIS (this §4 v0.1): operator-level realization of all 4 engine pieces + quasi-eigentone framework for Vol 16.

Cal #27 / honesty: §4 is curriculum content synthesizing rigorous engine results pedagogically. Not new derivation; pedagogical packaging.

Routed: → Casey: Vol 16 §4 ready for inclusion alongside §3 in the chapter draft. → Keeper: §3 + §4 = the operator-level dictionary + engine combined; ready for K-audit when convenient. → me: continuing v0.2 sweep — next T1 v0.2.

— Lyra, Vol 16 Chapter 1 §4 v0.1 (extends §3 v0.1 = Vol 16 chapter content v0.2). Engine 4-piece structure at operator level: multiplication (fusion, structure constants = substrate primaries N_c/N_c·n_C/N_c·g), coproduct (decay, grading-conserved), R-matrix (scattering, braiding-induced), negative part (antimatter, CPT = bar involution). Quasi-eigentone framework places SM stable/unstable classification at A_sub eigenstate level. β-decay E3 generalized to all SM decays via Green coproduct. Pedagogical synthesis of Saturday's engine work for the Vol 16 curriculum.